B Ajaysvin

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Education

Amrita Vishwa Vidyapeetham Chennai, Tamil Nadu
Bachelor of Technology in Automation and Robotics Engineering *2022 – 2026*

Shri Nehru Vidyalaya Coimbatore, Tamil Nadu
Higher Secondary Education *2020 – 2022*

Projects

Voice Guided Robot with Precise Angle Navigation

- Designed and developed a two-wheeled robot that responds to voice commands via a mobile Bluetooth app, enabling hands-free operation.
- Integrated ultrasonic sensors for real-time obstacle detection and automatic halting to prevent collisions.
- Implemented angle-based turning functionality (30°, 45°, 90°, 180°) followed by forward motion, allowing precise directional control using simple voice inputs.
- Added IR sensor-based line-following mode and "Follow Me" command feature for autonomous path tracking and user following.

Condition Monitoring on Wind Turbines Using OpenCV | *OpenCV, Python, ML, Drone*

- Deployed drones for aerial data acquisition to inspect wind turbine blades.
- Implemented OpenCV-based algorithms to detect corrosion, erosion, and lightning-induced damages.
- Enabled remote, high-precision monitoring of critical infrastructure components.

Keyhole Navigator | *Embedded Automation , IoT Basics , Arduino*

- Developed a motion-activated device to illuminate keyholes in the dark using a proximity sensor and LED lights.
- Integrated a proximity sensor to detect user presence and trigger LED lighting.
- Enhanced user convenience by improving nighttime accessibility without manual effort.

Technical Skills

Languages: Python, C/C++
Hardware: Raspberry pi, Arduino, ESP32, Ultrasonic Sensors, IR Sensors, Bluetooth Modules, IMU
Libraries: pandas, NumPy, Matplotlib, OpenCV
Prototyping : Rapid Prototyping, Hardware Prototyping, Sensor Integration
Designing: Fusion 360, AutoCAD, CAD Design, 3D Printing
Office Tools: Proficient in Presentation design(PowerPoint), Word, Excel

Areas Of Interest

- Internet of Things (IoT)
- Prototyping and Hardware Development
- Robotics and Automation
- Basic Machine Learning and Predictive Modeling
- Embedded Systems and Sensor Integration
- Sensor Fusion and Real-Time Data Processing
- Data Analysis and Visualization
- CAD Designing and 3D Printing

Achievements and Awards

Won 1st Place in the 24-Hour Hackathon at *Ingenium 4.0 – Tantrotsav 2025*

- Developed a Vehicle-to-Vehicle (V2V) communication system using ArUco markers for navigation, enabling accurate marker-based identification and real-time data exchange between vehicles.

Poster Presentation – International Conference on Additive Manufacturing Characterization, IIT Bombay

- Presented a poster at an international conference on a machine learning-based surrogate model for predicting distortions in lattice structures produced by laser powder bed fusion, aiming to enhance accuracy and efficiency in additive manufacturing.